Unit-2

Representation of Data and Register Transfer with Micro-Operations

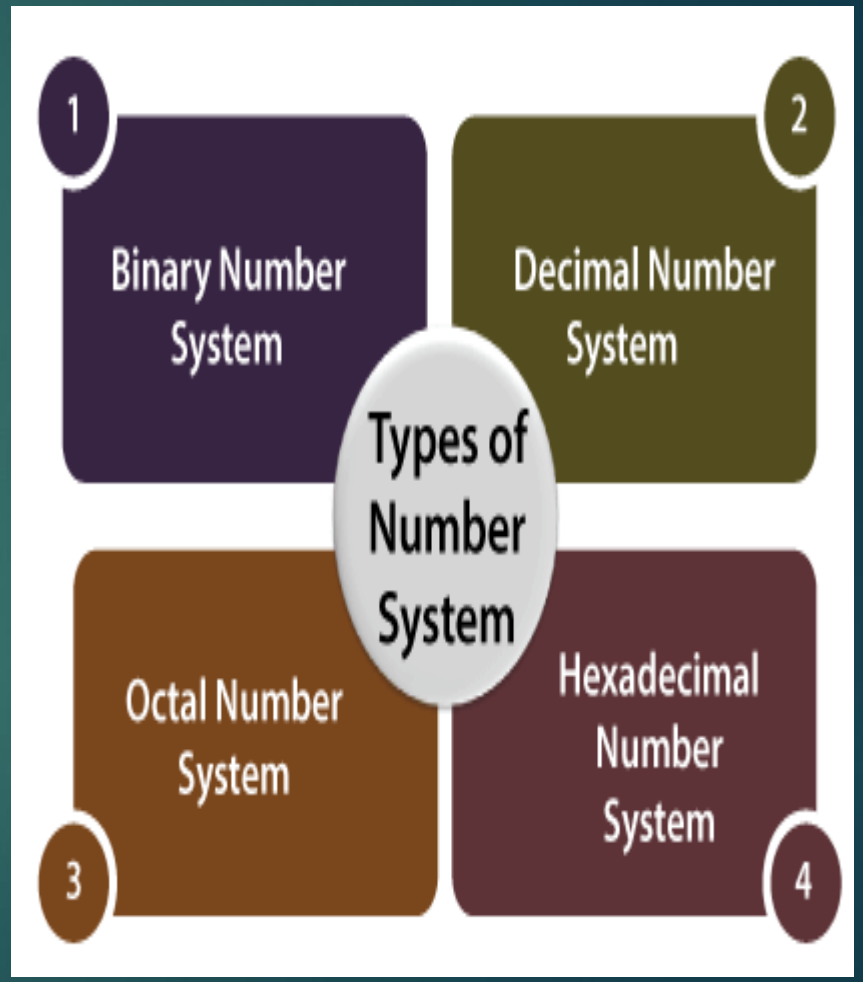
Data Type:

- Data Represents refers how to define data and store in computer.
- We all know that computer is used to data In binary format and we used data in higher level language ,
- so it is compulsory for computer to convert data in binary formation to store data.
- For storing a data in computer there are several different types are available to convert higher level data in to binary formats.
- There are several computational data representation in binary format.
- Represents data in signed format.
- Represent magnitude
- Decimal representation and fractional representation
- Exponent part of data

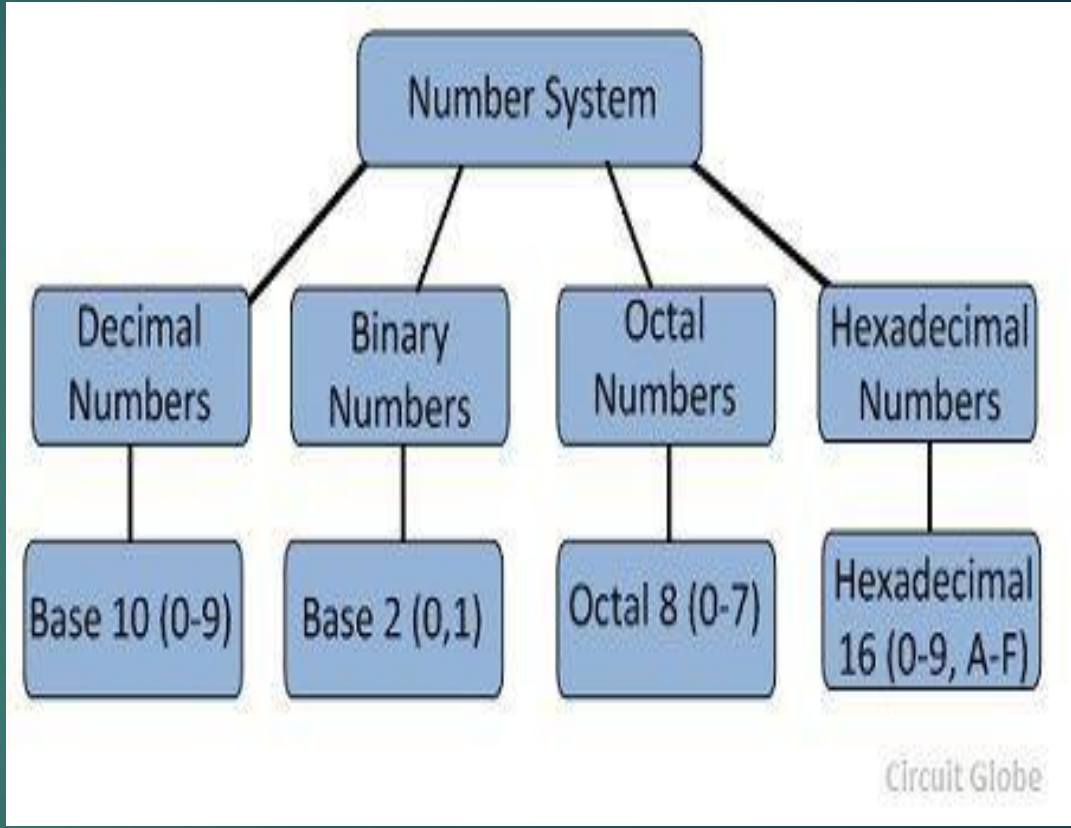
- In computer we give any data in different formats like character data, Number data, jpg, png, image data, Audio and video data.
- Each data are converted into binary formats by applying different techniques.
- For example, Suppose we write ‘A’ as a input the computer will store ‘A’ in binary format in memory.
- For that computer convert ‘A’ letter in to ASCII format.
- Here ASCII of A is 65
- So computer will now convert binary of 65 and store In memory,
- For conversion computer use different number system

Number System:

- Number system is used for such a conversion to convert data higher level to binary language.
- To a computer, everything is a number, i.e., alphabets, pictures, sounds, etc., are numbers.
- Number system is categorized into four types –
- Binary Number System
- Decimal Number System
- Hexadecimal Number System
- Octal Number System



- Binary number system consists of only two values, either 0 or 1
- Octal number system represents values in 8 digits.
- Decimal number system represents values in 10 digits.
- Hexadecimal number system represents values in 16 digits.



Binary Number System:

- Generally, a binary number system is used in the digital computers.
- In this number system, it carries only two digits, either 0 or 1.
- There are two types of electronic pulses present in a binary number system.
- The first one is the absence of an electronic pulse representing '0'and second one is the presence of electronic pulse representing '1'.
- Each digit is known as a bit.
- A four-bit collection (1101) is known as a nibble, and a collection of eight bits (11001010) is known as a byte.
- The location of a digit in a binary number represents a specific power of the base (2) of the number system.

- It holds only two values, i.e., either 0 or 1.
- It is also known as the base 2 number system.
- The position of a digit represents the 0 power of the base(2). Example: 2^0
- The position of the last digit represents the x power of the base(2). Example: 2^x, where x represents the last position, i.e., 1
- **Examples:**
- $(10100)_2$, $(11011)_2$, $(11001)_2$, $(000101)_2$, $(011010)_2$.

Decimal Number System

- The decimal numbers are used in our day to day life.
- The decimal number system contains ten digits from **0 to 9(base 10).** Here, the successive place value or position, left to the decimal point holds units, tens, hundreds, thousands, and so on.
- The position in the decimal number system specifies the power of the base (10).
- The 0 is the minimum value of the digit, and 9 is the maximum value of the digit.
- For example, the decimal number 2541 consist of the digit 1 in the unit position, 4 in the tens position, 5 in the hundreds position, and 2 in the thousand positions and the value will be written as:

- $(2\times1000) + (5\times100) + (4\times10) + (1\times1)$
- $2000 + 500 + 40 + 1$
- 2541

Octal Number System:

- The octal number system has **base 8**(means it has only eight digits from 0 to 7).
- There are only eight possible digit values to represent a number.
- With the help of only three bits, an octal number is represented.
- Each set of bits has a distinct value between 0 and 7.
- Below, we have described certain characteristics of the octal number system:
- **Characteristics:**
- An octal number system carries eight digits starting from 0, 1, 2, 3, 4, 5, 6, and 7.
- It is also known as the base 8 number system.
- The position of a digit represents the 0 power of the base(8). Example: 8^0
- The position of the last digit represents the x power of the base(8). Example: 8^x, where x represents the last position, i.e., 1
- **Examples:**
- $(273)_8$, $(5644)_8$, $(0.5365)_8$, $(1123)_8$, $(1223)_8$.

Number	Octal Number
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Hexadecimal Number System:

- It is another technique to represent the number in the digital system called the **hexadecimal number system**.
- The number system has a base of 16 means there are total 16 symbols(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F) used for representing a number.
- The single-bit representation of decimal values10, 11, 12, 13, 14, and 15 are represented by A, B, C, D, E, and F.
- Only 4 bits are required for representing a number in a hexadecimal number.
- Each set of bits has a distinct value between 0 and 15. There are the following characteristics of the octal number system:
- **Characteristics:**
- It has ten digits from 0 to 9 and 6 letters from A to F.
- The letters from A to F defines numbers from 10 to 15.
- It is also known as the base 16number system.
- In hexadecimal number, the position of a digit represents the 0 power of the base(16). Example: 16^0
- In hexadecimal number, the position of the last digit represents the x power of the base(16). Example: 16^x, where x represents the last position, i.e., 1
- $(FAC2)_{16}$, $(564)_{16}$, $(0ABD5)_{16}$, $(1123)_{16}$, $(11F3)_{16}$.

Binary Number	Hexadecimal Number
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

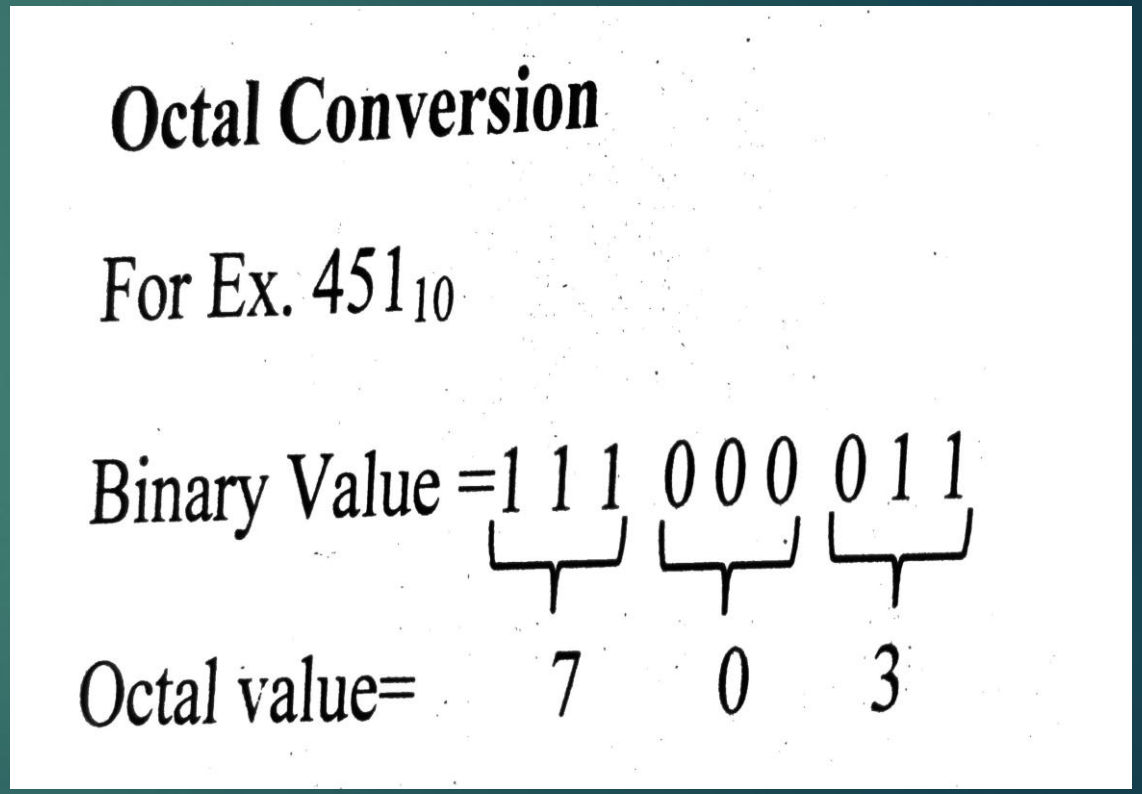
Conversion of Decimal Number into Binary , Octal and Hexadecimal

- **Binary conversion:**
- Ex: 451
- For converting a binary value will used LCM technique for that we will modulo the decimal number by 2.
- Now write reminder part from bottom to top it will be a binary number of your decimal number.
- Binary number= 111000011

2	451	
2	225	1
2	112	1
2	56	0
2	28	0
2	14	0
2	7	0
2	3	1
2	1	1
2	0	1

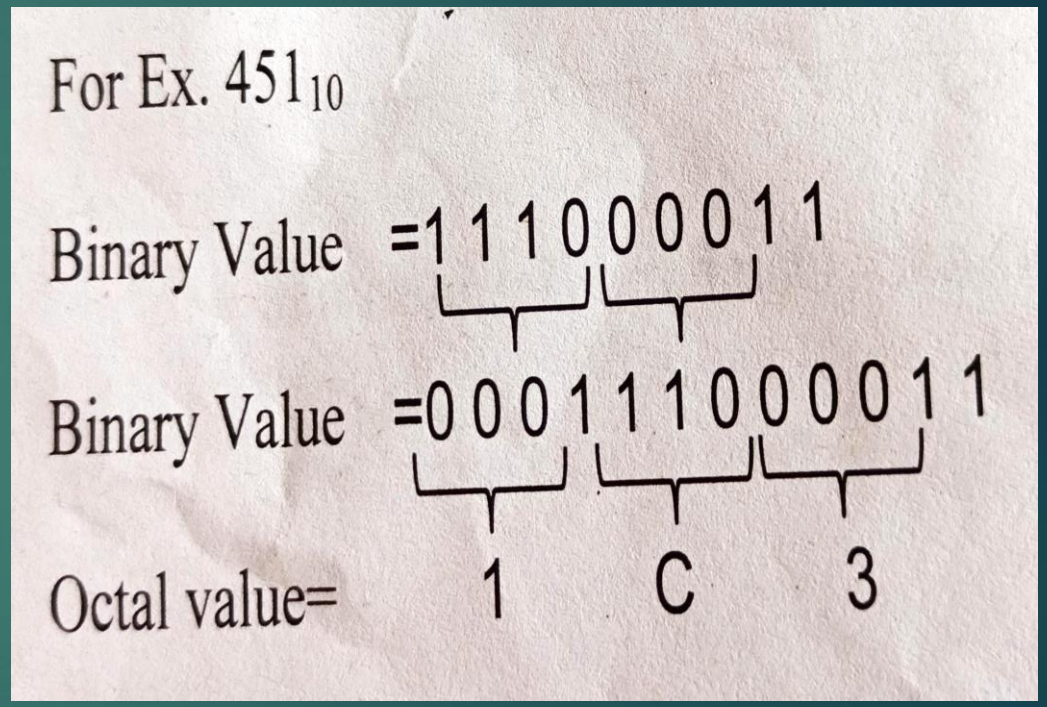
Octal Conversion:

- For ex: 451_{10}
- For converting a octal value to , first of al converting into binary number from decimal number,
- Then make a bunch of 3 binary digits from right to left, then convert Binary to octal number.
- Octal value=703_8



Hexadecimal Conversion:

- Ex: 451_{10}
- For converting an hexadecimal value , first of all convert into binary .
- Then make a bunch of 4 binary digits from left to right
- Then convert Binary number into Hexadecimal Number.
- In this example, You find last digit has only single digit to make a bunch of 4, we will add 3 times 0 after the single digit 1.
- Hexadecimal Value:1C3



Alpha Numeric Representation:

- ASCII represents **American Standard Code for Information Interchange**.
- It is the standard binary code used to represent alphanumeric characters.
- Alphanumeric characters are used for the transfer of information to and from the I/O devices and the computer.
- This standard helps seven bits to code 128 characters. However, there is an additional bit on the left that is always assigned 0. Therefore, there are 8 bits in total.
- The ASCII code consists of 34 nonprinting characters and 94 characters used for various control operations. There are 26 uppercase letters A through Z, 26 lowercase letters a through z, numerals from 0 to 9, and 32 printable characters including %,*.
- The control characters are used to route the data and arrange the printed text into a prescribed format.

TABLE 3-4 American Standard Code for Information Interchange (ASCII)

Character	Binary code	Character	Binary code
A	100 0001	0	011 0000
B	100 0010	1	011 0001
C	100 0011	2	011 0010
D	100 0100	3	011 0011
E	100 0101	4	011 0100
F	100 0110	5	011 0101
G	100 0111	6	011 0110
H	100 1000	7	011 0111
I	100 1001	8	011 1000
J	100 1010	9	011 1001
K	100 1011		
L	100 1100		
M	100 1101	space	010 0000
N	100 1110	.	010 1110
O	100 1111	(	010 1000
P	101 0000	+	010 1011
Q	101 0001	$	010 0100
R	101 0010	*	010 1010
S	101 0011	)	010 1001
T	101 0100	—	010 1101
U	101 0101	/	010 1111
V	101 0110	,	010 1100
W	101 0111	=	011 1101
X	101 1000		
Y	101 1001		
Z	101 1010		